

Technical Documentation

RAG-Based Customer Support Assistant

1. Introduction

Retrieval-Augmented Generation (RAG) combines document retrieval with AI-based answer generation to improve accuracy by using real data from a knowledge base.

This system acts as a customer support assistant that answers queries using a PDF knowledge base.

2. System Architecture

PDF → Chunking → Embeddings → ChromaDB → Query → Retrieval → LLM → Answer

3. Design Decisions

Chunk size is around 500 to balance context and performance. Sentence Transformers are used for embeddings. Top-k retrieval is used for fetching relevant chunks.

4. Workflow (LangGraph)

Input Node → Processing Node → Output Node

The system retrieves context, processes queries, and generates answers using a graph-based workflow.

5. Conditional Logic

If confidence is low or no data is found, the query is escalated. Otherwise, the system generates an answer.

6. Human-in-the-Loop (HITL)

HITL handles complex queries by forwarding them to a human, who provides the final response.

7. Challenges and Trade-offs

Balancing accuracy and speed, managing chunk size, and optimizing cost vs performance are key challenges.

8. Testing

Tested using normal, unknown, and complex queries to validate system behavior.

9. Future Enhancements

Support multiple documents, add chat memory, and develop a web interface.

10. Mandatory Concepts Coverage

This project includes RAG, PDF ingestion, chunking, embeddings with ChromaDB, retrieval, LangGraph workflow, Input-Process-Output flow, conditional routing, customer support use case, and HITL escalation.